
A Summations

When an algorithm contains an iterative control construct such as a **while** or **for** loop, you can express its running time as the sum of the times spent on each execution of the body of the loop. For example, Section 2.2 argued that the i th iteration of insertion sort took time proportional to i in the worst case. Adding up the time spent on each iteration produced the summation (or series) $\sum_{i=2}^n i$. Evaluating this summation resulted in a bound of $\Theta(n^2)$ on the worst-case running time of the algorithm. This example illustrates why you should know how to manipulate and bound summations.

Section A.1 lists several basic formulas involving summations. Section A.2 offers useful techniques for bounding summations. The formulas in Section A.1 appear without proof, though proofs for some of them appear in Section A.2 to illustrate the methods of that section. You can find most of the other proofs in any calculus text.

A.1 Summation formulas and properties

Given a sequence $a_1, a_2, \dots, a_n$ of numbers, where n is a nonnegative integer, the finite sum $a_1 + a_2 + \dots + a_n$ can be expressed as $\sum_{k=1}^n a_k$. If $n = 0$, the value of the summation is defined to be 0. The value of a finite series is always well defined, and the order in which its terms are added does not matter.

Given an infinite sequence $a_1, a_2, \dots$ of numbers, we can write their infinite sum $a_1 + a_2 + \dots$ as $\sum_{k=1}^{\infty} a_k$, which means $\lim_{n \rightarrow \infty} \sum_{k=1}^n a_k$. If the limit does not exist, the series *diverges*, and otherwise, it *converges*. The terms of a convergent series cannot always be added in any order. You can, however, rearrange the terms of an *absolutely convergent series*, that is, a series $\sum_{k=1}^{\infty} a_k$ for which the series $\sum_{k=1}^{\infty} |a_k|$ also converges.

Linearity

For any real number c and any finite sequences $a_1, a_2, \dots, a_n$ and $b_1, b_2, \dots, b_n$,

$$\sum_{k=1}^n (ca_k + b_k) = c \sum_{k=1}^n a_k + \sum_{k=1}^n b_k .$$

The linearity property also applies to infinite convergent series.

The linearity property applies to summations incorporating asymptotic notation. For example,

$$\sum_{k=1}^n \Theta(f(k)) = \Theta\left(\sum_{k=1}^n f(k)\right) .$$

In this equation, the Θ -notation on the left-hand side applies to the variable k , but on the right-hand side, it applies to n . Such manipulations also apply to infinite convergent series.

Arithmetic series

The summation

$$\sum_{k=1}^n k = 1 + 2 + \dots + n ,$$

is an *arithmetic series* and has the value

$$\sum_{k=1}^n k = \frac{n(n+1)}{2} \tag{A.1}$$

$$= \Theta(n^2) . \tag{A.2}$$

A *general arithmetic series* includes an additive constant $a \geq 0$ and a constant coefficient $b > 0$ in each term, but has the same total

asymptotically:

$$\sum_{k=1}^n (a + bk) = \Theta(n^2) . \quad (\text{A.3})$$

Sums of squares and cubes

The following formulas apply to summations of squares and cubes:

$$\sum_{k=0}^n k^2 = \frac{n(n+1)(2n+1)}{6} , \quad (\text{A.4})$$

$$\sum_{k=0}^n k^3 = \frac{n^2(n+1)^2}{4} . \quad (\text{A.5})$$

Geometric series

For real $x \neq 1$, the summation

$$\sum_{k=0}^n x^k = 1 + x + x^2 + \cdots + x^n$$

is a *geometric series* and has the value

$$\sum_{k=0}^n x^k = \frac{x^{n+1} - 1}{x - 1} . \quad (\text{A.6})$$

The infinite decreasing geometric series occurs when the summation is infinite and $|x| < 1$:

$$\sum_{k=0}^{\infty} x^k = \frac{1}{1 - x} . \quad (\text{A.7})$$

If we assume that $0^0 = 1$, these formulas apply even when $x = 0$.

Harmonic series

For positive integers n , the n th *harmonic number* is

$$\begin{aligned} H_n &= 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots + \frac{1}{n} \\ &= \sum_{k=1}^n \frac{1}{k} \end{aligned} \quad (\text{A.8})$$

$$= \ln n + O(1) . \quad (\text{A.9})$$

Inequalities (A.20) and (A.21) on page 1150 provide the stronger bounds

$$\ln(n+1) \leq H_n \leq \ln n + 1. \quad (\text{A.10})$$

Integrating and differentiating series

Integrating or differentiating the formulas above yields additional formulas. For example, differentiating both sides of the infinite geometric series (A.7) and multiplying by x gives

$$\sum_{k=0}^{\infty} kx^k = \frac{x}{(1-x)^2} \quad (\text{A.11})$$

for $|x| < 1$.

Telescoping series

For any sequence $a_0, a_1, \dots, a_n$,

$$\sum_{k=1}^n (a_k - a_{k-1}) = a_n - a_0, \quad (\text{A.12})$$

since each of the terms $a_1, a_2, \dots, a_{n-1}$ is added in exactly once and subtracted out exactly once. We say that the sum *telescopes*. Similarly,

$$\sum_{k=0}^{n-1} (a_k - a_{k+1}) = a_0 - a_n.$$

As an example of a telescoping sum, consider the series

$$\sum_{k=1}^{n-1} \frac{1}{k(k+1)}.$$

Rewriting each term as

$$\frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1},$$

gives

$$\begin{aligned}\sum_{k=1}^{n-1} \frac{1}{k(k+1)} &= \sum_{k=1}^{n-1} \left(\frac{1}{k} - \frac{1}{k+1} \right) \\ &= 1 - \frac{1}{n}.\end{aligned}$$

Reindexing summations

A series can sometimes be simplified by changing its index, often reversing the order of summation. Consider the series $\sum_{k=0}^n a_{n-k}$. Because the terms in this summation are $a_n, a_{n-1}, \dots, a_0$, we can reverse the order of indices by letting $j = n - k$ and rewrite this summation as

$$\sum_{k=0}^n a_{n-k} = \sum_{j=0}^n a_j. \quad (\text{A.13})$$

Generally, if the summation index appears in the body of the sum with a minus sign, it's worth thinking about reindexing.

As an example, consider the summation

$$\sum_{k=1}^n \frac{1}{n-k+1}.$$

The index k appears with a negative sign in $1/(n-k+1)$. And indeed, we can simplify this summation, this time setting $j = n - k + 1$, yielding

$$\sum_{k=1}^n \frac{1}{n-k+1} = \sum_{j=1}^n \frac{1}{j}, \quad (\text{A.14})$$

which is just the harmonic series (A.8).

Products

The finite product $a_1 a_2 \dots a_n$ can be expressed as

$$\prod_{k=1}^n a_k.$$

If $n = 0$, the value of the product is defined to be 1. You can convert a formula with a product to a formula with a summation by using the identity

$$\lg \left(\prod_{k=1}^n a_k \right) = \sum_{k=1}^n \lg a_k .$$

Exercises

A.1-1

Prove that $\sum_{k=1}^n O(f_k(i)) = O(\sum_{k=1}^n f_k(i))$ by using the linearity property of summations.

A.1-2

Find a simple formula for $\sum_{k=1}^n (2k-1)$.

A.1-3

Interpret the decimal number 111,111,111 in light of equation (A.6).

A.1-4

Evaluate the infinite series $1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \frac{1}{16} - \dots$.

A.1-5

Let $c \geq 0$ be a constant. Show that $\sum_{k=1}^n k^c = \Theta(n^{c+1})$.

A.1-6

Show that $\sum_{k=0}^{\infty} k^2 x^k = x(1+x)/(1-x)^3$ for $|x| < 1$.

A.1-7

Prove that $\sum_{k=1}^n \sqrt{k \lg k} = \Theta(n^{3/2} \lg^{1/2} n)$. (*Hint: Show the asymptotic upper and lower bounds separately.*)

★ A.1-8

Show that $\sum_{k=1}^n 1/(2k-1) = \ln(\sqrt{n}) + O(1)$ by manipulating the harmonic series.

★ A.1-9

Show that $\sum_{k=0}^{\infty} (k-1)/2^k = 0$.

★ A.1-10

Evaluate the sum $\sum_{k=1}^{\infty} (2k+1)x^{2k}$ for $|x| < 1$.

★ A.1-11

Evaluate the product $\prod_{k=2}^n (1 - 1/k^2)$.

A.2 Bounding summations

You can choose from several techniques to bound the summations that describe the running times of algorithms. Here are some of the most frequently used methods.

Mathematical induction

The most basic way to evaluate a series is to use mathematical induction. As an example, let's prove that the arithmetic series $\sum_{k=1}^n k$ evaluates to $n(n+1)/2$. For $n = 1$, we have that $n(n+1)/2 = 1 \cdot 2/2 = 1$, which equals $\sum_{k=1}^1 k$. With the inductive assumption that it holds for n , we prove that it holds for $n+1$. We have

$$\begin{aligned}\sum_{k=1}^{n+1} k &= \sum_{k=1}^n k + (n+1) \\ &= \frac{n(n+1)}{2} + (n+1) \\ &= \frac{n^2 + n + 2n + 2}{2} \\ &= \frac{(n+1)(n+2)}{2}.\end{aligned}$$

You don't always need to guess the exact value of a summation in order to use mathematical induction. Instead, you can use induction to prove an upper or lower bound on a summation. As an example, let's prove the asymptotic upper bound $\sum_{k=0}^n 3^k = O(3^n)$. More specifically, we'll prove that $\sum_{k=0}^n 3^k \leq c3^n$ for some constant c . For the initial condition $n = 0$, we have $\sum_{k=0}^0 3^k = 1 \leq c$ as long as $c \geq 1$. Assuming that the bound holds for n , we prove that it holds for $n+1$. We have

$$\begin{aligned}\sum_{k=0}^{n+1} 3^k &= \sum_{k=0}^n 3^k + 3^{n+1} \\ &\leq c3^n + 3^{n+1} \quad (\text{by the inductive hypothesis}) \\ &= \left(\frac{1}{3} + \frac{1}{c}\right) c3^{n+1} \\ &\leq c3^{n+1}\end{aligned}$$